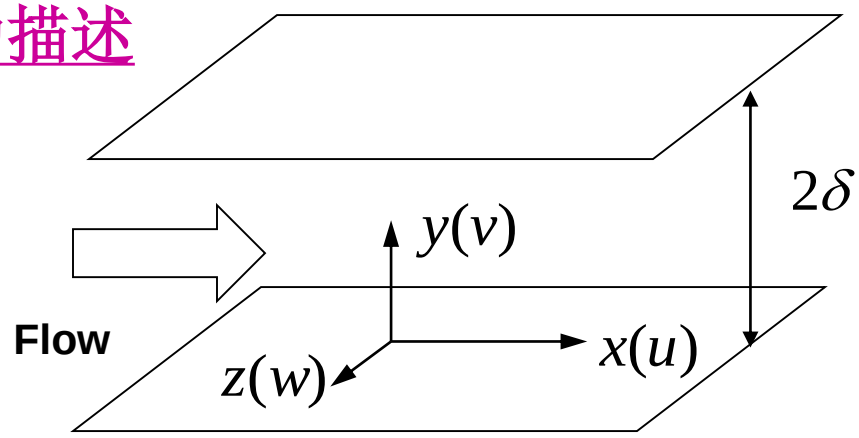
§6.4 壁湍流的统计特性

一、槽道湍流

流动描述



$y=2\delta$ 上壁面

$y=\delta$ 中心线

$y=0$ 下壁面

$$\frac{\partial\langle\ \rangle}{\partial t}=0 \quad \frac{\partial\langle\ \rangle}{\partial x}=0\text{（压强除外）} \quad \frac{\partial\langle\ \rangle}{\partial z}=0$$

$$\langle u\rangle=U(y),\quad \langle v\rangle=0,\quad \langle w\rangle=0 \qquad \langle v'w'\rangle=\langle u'w'\rangle=0$$

$$Re_m=\frac{2U_m\delta}{\nu} \qquad Re_c=\frac{U_c\delta}{\nu} \qquad U_m=\frac{1}{2\delta}\int_0^{2\delta}U(y)\mathrm{d}y \qquad U_c=U(\delta)$$

§6.4 壁湍流的统计特性

平均运动方程

$$\frac{\partial \langle V_i \rangle}{\partial t} + \langle V_j \rangle \frac{\partial \langle V_i \rangle}{\partial x_j} = -\frac{1}{\rho}\frac{\partial \langle p \rangle}{\partial x_i} + \nu \frac{\partial^2 \langle V_i \rangle}{\partial x_j \partial x_j} - \frac{\partial}{\partial x_j}\langle V_i' V_j' \rangle \qquad \frac{\partial \langle V_i \rangle}{\partial x_i} = 0$$

$$0 = -\frac{1}{\rho}\frac{\partial \langle p \rangle}{\partial x} + \nu \frac{\partial^2 U}{\partial y^2} - \frac{\partial}{\partial y}\langle u'v' \rangle$$

$$0 = -\frac{1}{\rho}\frac{\partial \langle p \rangle}{\partial y} - \frac{\partial}{\partial y}\langle v'^2 \rangle \quad \Rightarrow \quad \frac{\langle p \rangle}{\rho} + \langle v'^2 \rangle = \frac{p_w(x)}{\rho} \quad \Rightarrow \quad \frac{\partial \langle p \rangle}{\partial x} = \frac{\mathrm{d}p_w}{\mathrm{d}x}$$

$$0 = -\frac{1}{\rho}\frac{\partial \langle p \rangle}{\partial z}$$

$$\frac{1}{\rho}\frac{\mathrm{d}p_w}{\mathrm{d}x} = \nu \frac{\partial^2 U}{\partial y^2} - \frac{\partial}{\partial y}\langle u'v' \rangle \quad \Rightarrow \quad \boxed{\mu \frac{\partial U}{\partial y} - \rho \langle u'v' \rangle = \tau_w \left(1 - \frac{y}{\delta}\right)} \qquad \tau_w = -\frac{\mathrm{d}p_w}{\mathrm{d}x}\delta$$

总切应力与离开壁面的距离成线性关系

§6.4 壁湍流的统计特性

平均速度剖面

充分发展的槽道湍流

$$\mu\frac{\mathrm{d}U}{\mathrm{d}y}-\rho\langle u'v'\rangle=\tau_w\left(1-\frac{y}{\delta}\right)$$

$$\tau_w\equiv\mu\left(\frac{\mathrm{d}U}{\mathrm{d}y}\right)_w=-\frac{\mathrm{d}p_w}{\mathrm{d}x}\delta=\rho u_\tau^2$$

1. 极近壁区

摩擦速度 $u_\tau\equiv\sqrt{\dfrac{\tau_w}{\rho}}$

粘性应力占主导，$y/\delta\ll1$

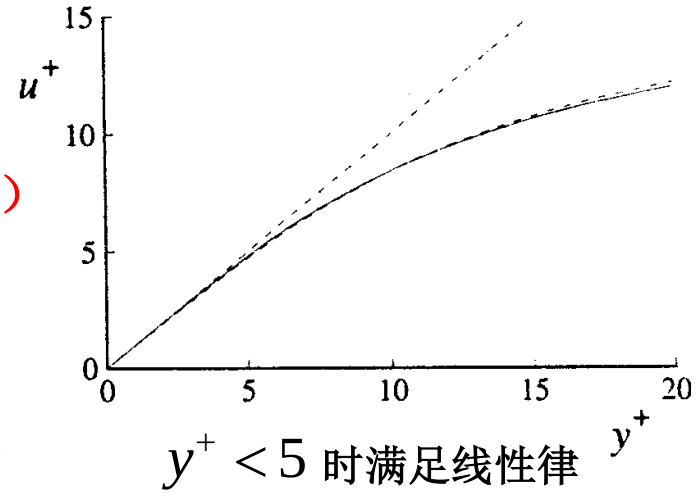
$$\mu\frac{\mathrm{d}U}{\mathrm{d}y}=\tau_w=\rho u_\tau^2$$

$$\Longrightarrow\quad U=\frac{u_\tau^2}{\nu}y \quad 或 \quad U^+=y^+ \quad（线性律）$$

$$U^+=\frac{U}{u_\tau}\qquad y^+=\frac{yu_\tau}{\nu}=\frac{y}{\delta_\nu}\qquad \delta_\nu=\frac{\nu}{u_\tau}$$

局部雷诺数
惯性 / 粘性量级

壁面粘性尺度



$y^+<5$ 时满足线性律

§6.4 壁湍流的统计特性

$$\mu\frac{\mathrm{d}U}{\mathrm{d}y}-\rho\langle u'v'\rangle=\tau_w\left(1-\frac{y}{\delta}\right)$$

2. 近壁区

雷诺应力占主导，雷诺数足够高时 $y/\delta\ll 1,\ y^+\gg 1$

Prandtl混合长度模式

$$\tau_{xy}=-\rho\langle u'v'\rangle=\rho l^2\left(\frac{\mathrm{d}U}{\mathrm{d}y}\right)^2$$

$$\langle u'v'\rangle=\tau_w/\rho=u_\tau^2 \qquad \text{等应力层}$$

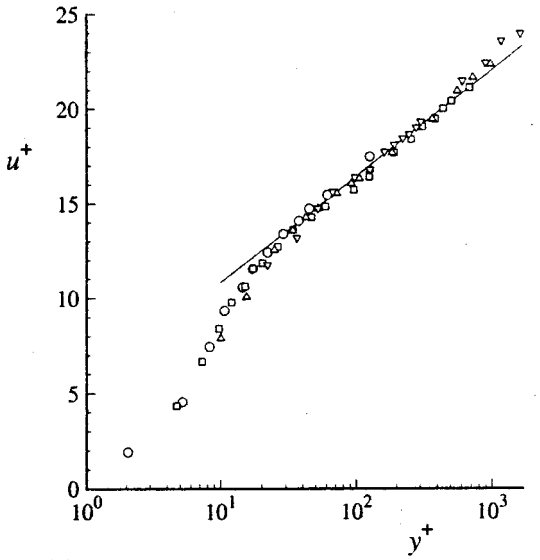
$$l=\kappa y$$

$$\Rightarrow\quad \frac{\mathrm{d}U}{\mathrm{d}y}=\frac{u_\tau}{\kappa y}$$

$$\Rightarrow\quad U=\frac{u_\tau}{\kappa}\ln y+B_1 \qquad \text{或}$$

$$U^+=\frac{1}{\kappa}\ln y^++B \qquad \text{（对数律）}$$

$\kappa=0.41$ 卡门常数　　$B=5.2$



结果证明：在 $y^+>30$ 且 $\bar{y}<0.3$ 区域，对数律成立。

§6.4 壁湍流的统计特性

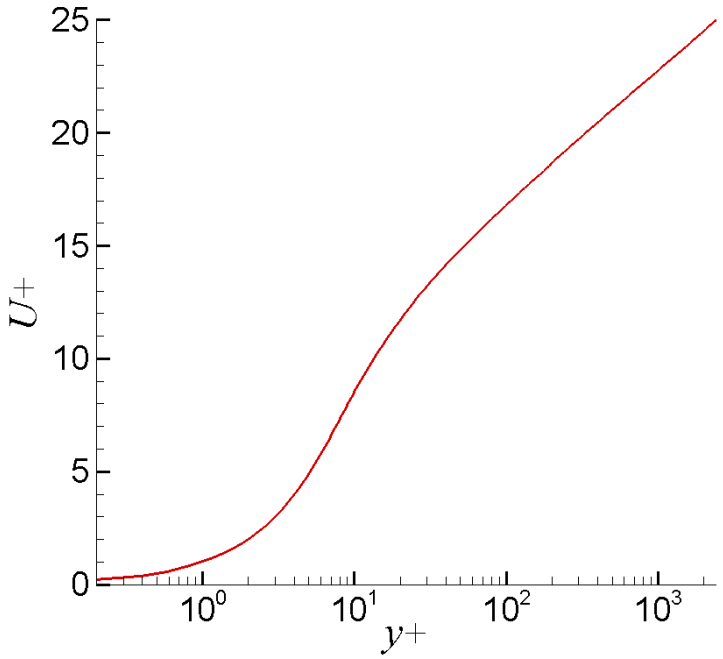
3. Spalding壁面律

粘性底层：$y^+ < 5,\ U^+ = y^+$

缓冲区：　？

对数区：$y^+ > 30, U^+ = \frac{1}{\kappa}\ln y^+ + B$

$$\Rightarrow \quad y^+ = e^{\kappa\left(U^+ - B\right)}$$



$$\Rightarrow \quad y^+ = U^+ + e^{-\kappa B}\left(e^{\kappa U^+} - 1 - \kappa U^+ - \left(\kappa U^+\right)^2/2 - \left(\kappa U^+\right)^3/6\right)$$

用途：1. 生成初始湍流平均速度剖面

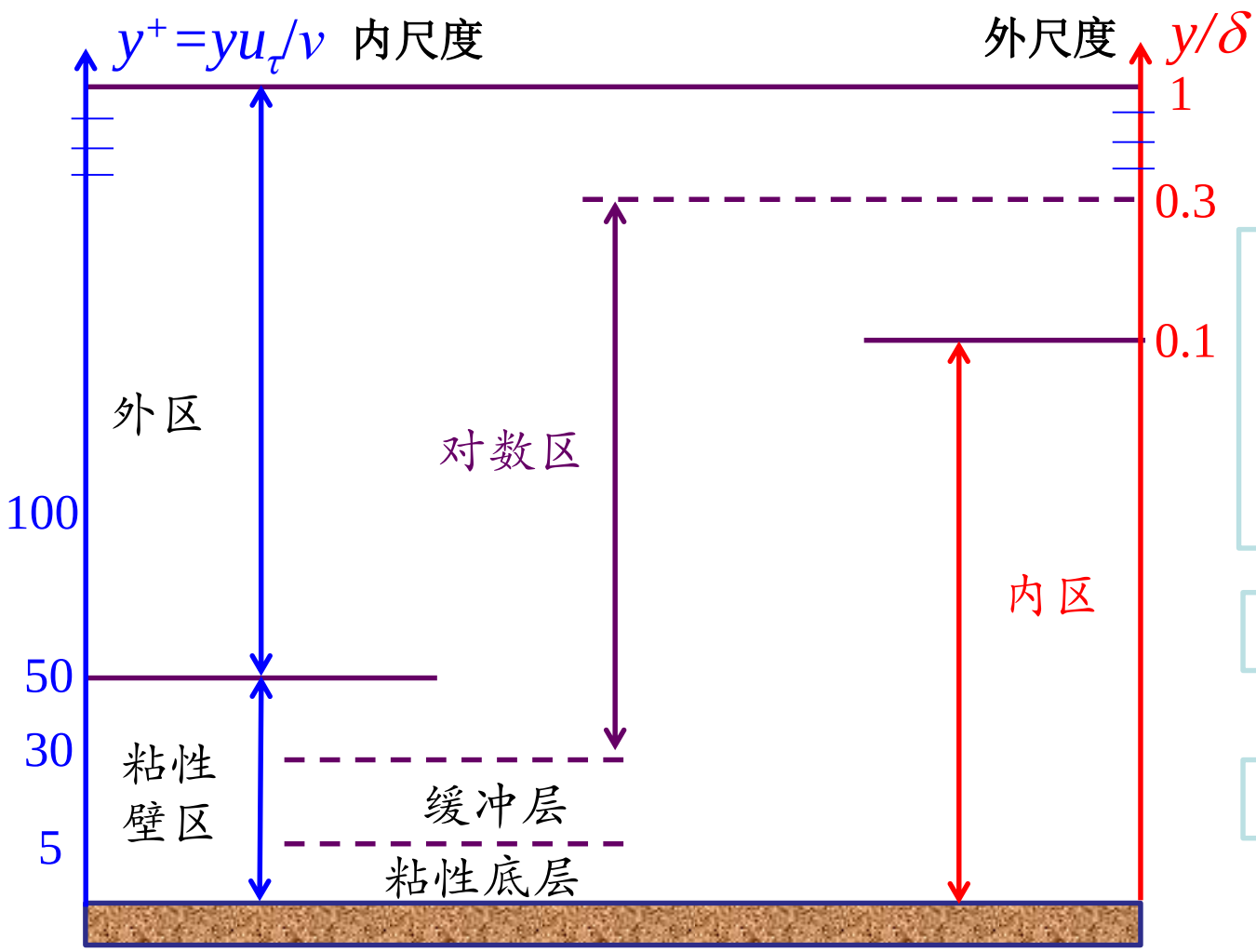
2. 估算涡粘系数：$dy^+/dU^+ = \nu_T/\nu$　　（**推导！**）

§6.4 壁湍流的统计特性

壁湍流的分层模型

$$y^+ = \bar{y} Re_\tau$$

$$\mathrm{Re}_\tau \equiv \frac{u_\tau \delta}{\nu} = \frac{\delta}{\delta_\nu}$$



大尺度相干结构
超大尺度相干结构
比如：
发卡涡包，
大尺度条带结构

发卡涡

条带结构，流向涡

§6.4 壁湍流的统计特性

雷诺应力对摩擦系数的贡献

$$\mu\frac{\mathrm{d}U}{\mathrm{d}y}-\rho\langle u'v'\rangle=\tau_w\left(1-\frac{y}{\delta}\right)\qquad \tau_w\equiv\mu\left(\frac{\mathrm{d}U}{\mathrm{d}y}\right)_w=-\frac{\mathrm{d}p_w}{\mathrm{d}x}\delta=\rho u_\tau^2$$

$$\int_0^\delta\int_0^y(\ )\,dydy$$

$$\Longrightarrow\quad \delta\mu U_m+\int_0^\delta(\delta-y)\left(-\rho\langle u'v'\rangle\right)dy=\frac{\delta^2}{3}\tau_w$$

取特征尺度：δ，$2U_m$

$$\Longrightarrow\quad C_f=\frac{\tau_w}{\frac{1}{2}\rho U_m^2}=\frac{12}{\mathrm{Re}}+24\int_0^1(1-\eta)\langle -u'v'\rangle\,d\eta$$

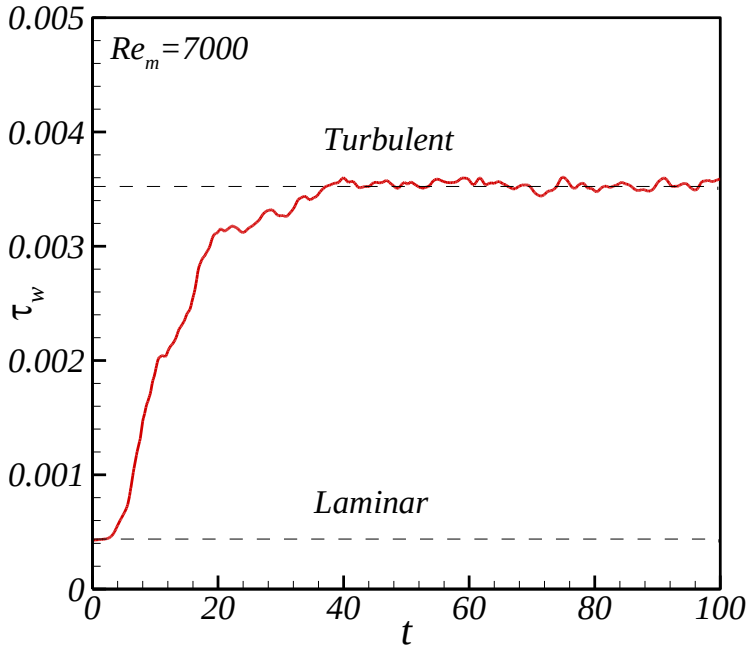
（推导！）

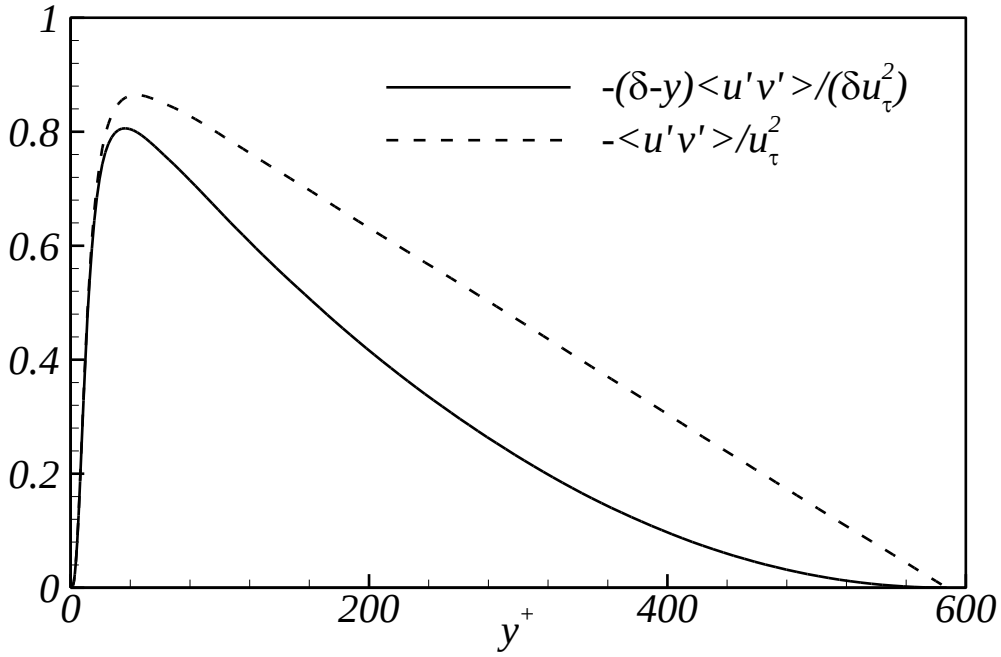
（FIK公式）

雷诺应力对摩擦系数的贡献随离壁面距离的增加而线性衰减，因此近壁处的雷诺应力的作用最为关键！

§6.4 壁湍流的统计特性

FIK公式：$C_f = \dfrac{\tau_w}{\frac{1}{2}\rho U_m^2} = \dfrac{12}{\mathrm{Re}} + 24\int_0^1 (1-\eta)\langle -u'v' \rangle d\eta$





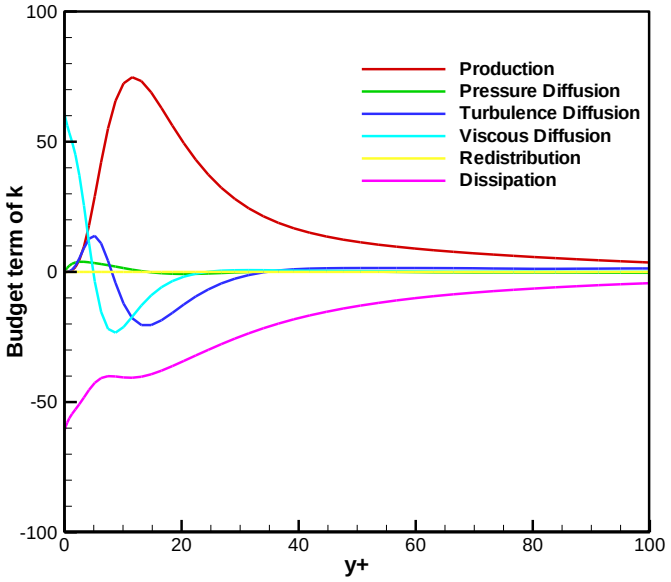
§6.4 壁湍流的统计特性

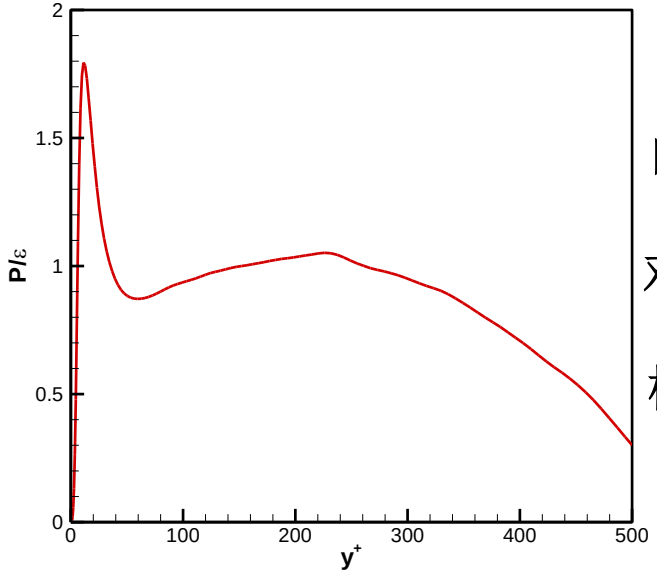
槽道湍流湍动能输运方程的平衡关系

$$\frac{\bar{D}k}{Dt} = P_k + D_k - \varepsilon$$

$$P_k = -\langle u'v' \rangle \frac{\mathrm{d}U}{\mathrm{d}y} \qquad D_k = \frac{\mathrm{d}}{\mathrm{d}y}\left(-\langle k'v' \rangle - \frac{1}{\rho}\langle p'v' \rangle + \nu \frac{\mathrm{d}k}{\mathrm{d}y} \right)$$

$$\varepsilon = \nu \left\langle \left(\frac{\partial u'}{\partial x}\right)^2 + \left(\frac{\partial u'}{\partial y}\right)^2 + \left(\frac{\partial u'}{\partial z}\right)^2 + \left(\frac{\partial v'}{\partial x}\right)^2 + \left(\frac{\partial v'}{\partial y}\right)^2 + \left(\frac{\partial v'}{\partial z}\right)^2 + \left(\frac{\partial w'}{\partial x}\right)^2 + \left(\frac{\partial w'}{\partial y}\right)^2 + \left(\frac{\partial w'}{\partial z}\right)^2 \right\rangle$$





内区： $P > \varepsilon$

对数区： $P \approx \varepsilon$

核心区： $P < \varepsilon$

湍流输运

§6.4 壁湍流的统计特性

局部各向同性湍流

对数区： $P \approx \varepsilon$

$$P_k = -\left\langle u'v' \right\rangle \frac{\mathrm{d}U}{\mathrm{d}y} = 2\nu_t S_{ij} S_{ij}$$

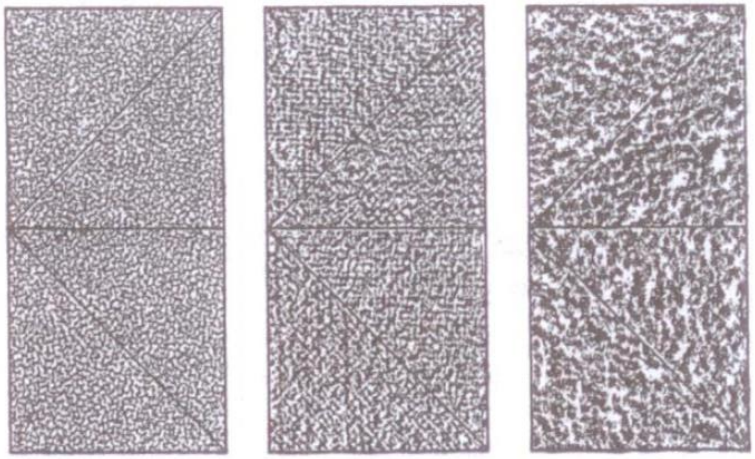
$$\varepsilon = 2\nu \left\langle s_{ij} s_{ij} \right\rangle$$

$\Longrightarrow$ $\left\langle s_{ij} s_{ij} \right\rangle \gg S_{ij} S_{ij}$

相对于流动时间尺度，对能量耗散有着大部分贡献的漩涡的时间尺度更小。

小尺度湍流结构往往独立于平均剪切引起的空间取向影响，所以所有与小尺度漩涡相关的平均值在坐标系旋转或者反射是都不发生变化。

这种情况下的小尺度湍流结构被称为各向同性。在小尺度上的各向同性被称为局部各向同性。



§6.4 壁湍流的统计特性

泰勒微尺度（λ）

$$\varepsilon = 2\nu\left\langle s_{ij}s_{ij}\right\rangle \sim \nu\frac{u'^2}{\lambda^2} \quad \Rightarrow \quad \lambda \sim \left(\frac{\nu u'^2}{\varepsilon}\right)^{\frac{1}{2}}$$

泰勒微尺度与大涡尺度之比：$\frac{\lambda}{l} \sim \mathrm{Re}^{-\frac{1}{2}}$

耗散涡的时间尺度：$\frac{\lambda}{u'} = \tau_\eta$

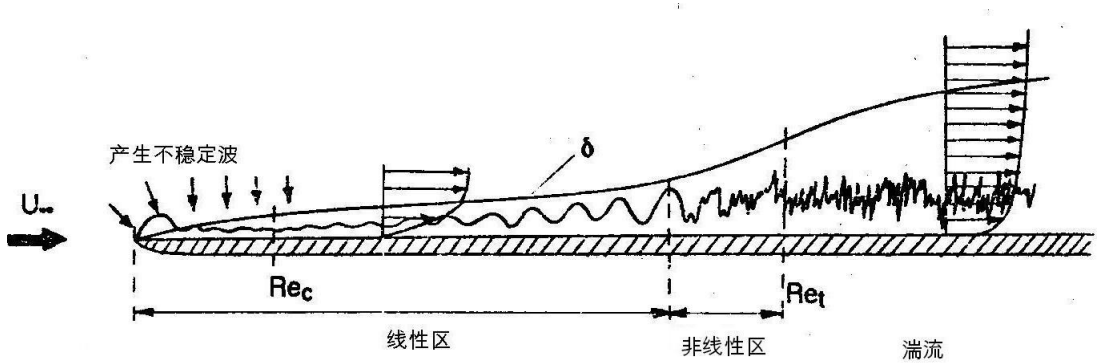
大涡的时间尺度：$\frac{l}{u'}$

泰勒雷诺数：$\mathrm{Re}_\lambda = \frac{u'\lambda}{\nu}$

§6.4 壁湍流的统计特性

二、湍流边界层

流动描述



$$\langle u\rangle = U(\varepsilon x, y),\quad \langle v\rangle = V \sim \varepsilon U,\quad \langle w\rangle = 0$$

$$\delta^*(x) \equiv \int_0^\infty \left(1-\frac{U}{U_e}\right)\mathrm{d}y \qquad \theta(x) \equiv \int_0^\infty \frac{U}{U_e}\left(1-\frac{U}{U_e}\right)\mathrm{d}y$$

$$Re_x = \frac{U_\infty x}{\nu} \qquad Re_{\delta^*} = \frac{U_\infty \delta^*}{\nu} \qquad Re_\theta = \frac{U_\infty \theta}{\nu}$$

§6.4 壁湍流的统计特性

平均运动方程

$$\frac{\partial \langle u_i \rangle}{\partial t} + \langle u_j \rangle \frac{\partial \langle u_i \rangle}{\partial x_j} = -\frac{1}{\rho}\frac{\partial \langle p \rangle}{\partial x_i} + \nu \frac{\partial^2 \langle u_i \rangle}{\partial x_j \partial x_j} - \frac{\partial}{\partial x_j}\langle u_i' u_j' \rangle$$

$$\frac{\partial \langle u_i \rangle}{\partial x_i} = 0$$

$$U\frac{\partial U}{\partial x} + V\frac{\partial U}{\partial y} = -\frac{1}{\rho}\frac{\partial P}{\partial x} + \nu\left(\frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2}\right) - \frac{\partial}{\partial x}\langle u'u' \rangle - \frac{\partial}{\partial y}\langle u'v' \rangle$$

$$\frac{\partial U}{\partial x} + \frac{\partial V}{\partial y} = 0$$

$$U\frac{\partial V}{\partial x} + V\frac{\partial V}{\partial y} = -\frac{1}{\rho}\frac{\partial P}{\partial y} + \nu\left(\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2}\right) - \frac{\partial}{\partial x}\langle u'v' \rangle - \frac{\partial}{\partial y}\langle v'v' \rangle$$

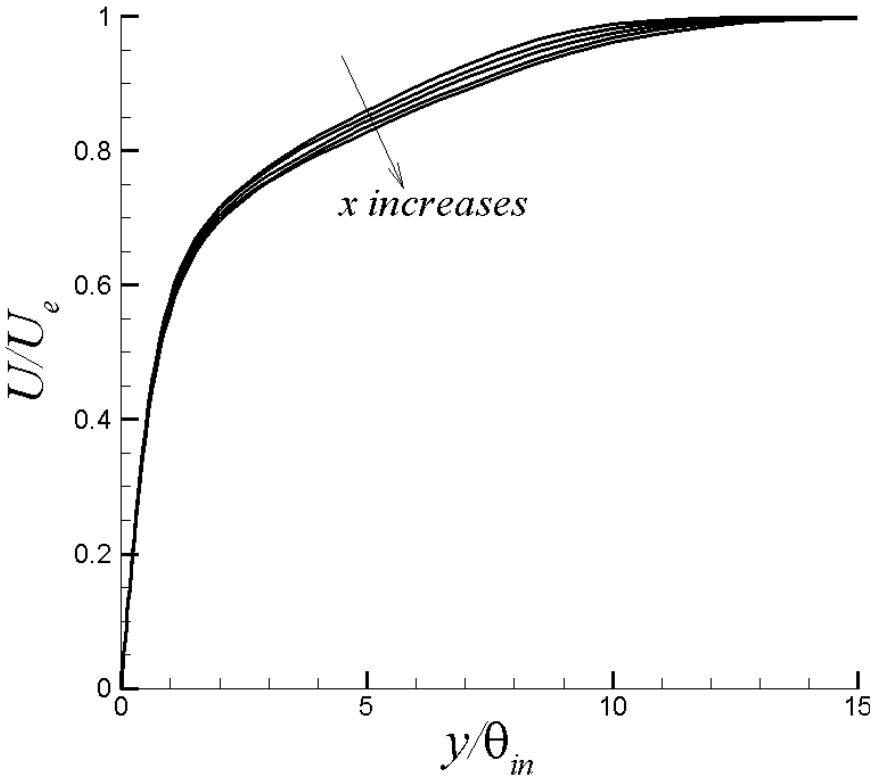
$$P + \rho\langle v'v' \rangle = P_\infty(x) = P_w(x) \Rightarrow \frac{\partial P}{\partial x} = \frac{\mathrm{d}P_\infty(x)}{\mathrm{d}x} - \rho\frac{\partial \langle v'v' \rangle}{\partial x}$$

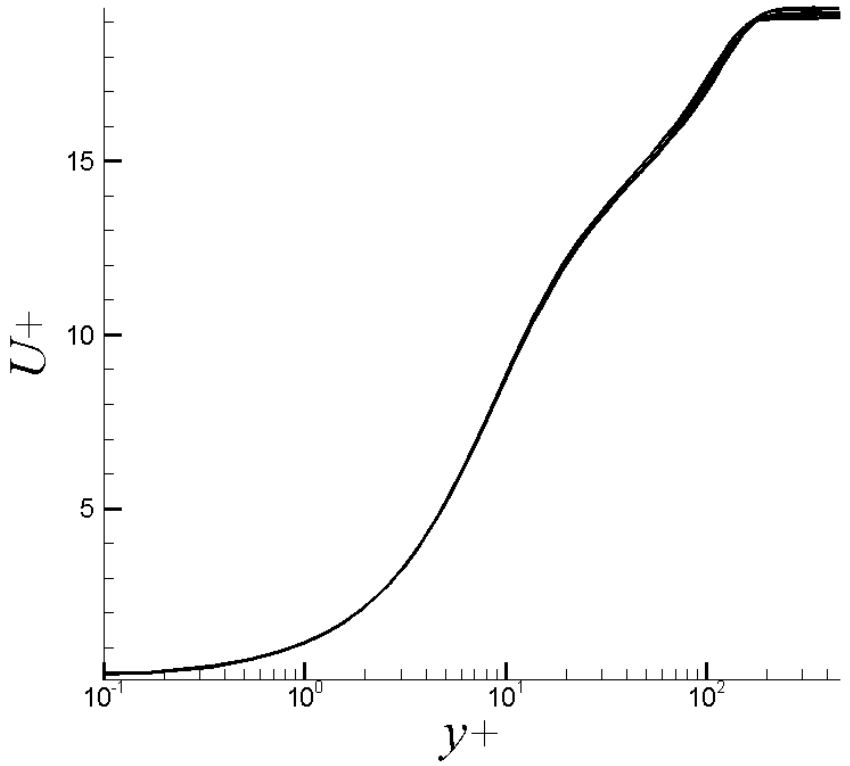
$$U\frac{\partial U}{\partial x} + V\frac{\partial U}{\partial y} = -\frac{1}{\rho}\frac{\mathrm{d}P_\infty}{\mathrm{d}x} + \nu\frac{\partial^2 U}{\partial y^2} - \frac{\partial}{\partial y}\langle u'v' \rangle$$

$$\Rightarrow \quad U\frac{\partial U}{\partial x} + V\frac{\partial U}{\partial y} = U_e\frac{dU_e}{dx} + \nu\frac{\partial^2 U}{\partial y^2} - \frac{\partial}{\partial y}\langle u'v' \rangle$$

§6.4 壁湍流的统计特性

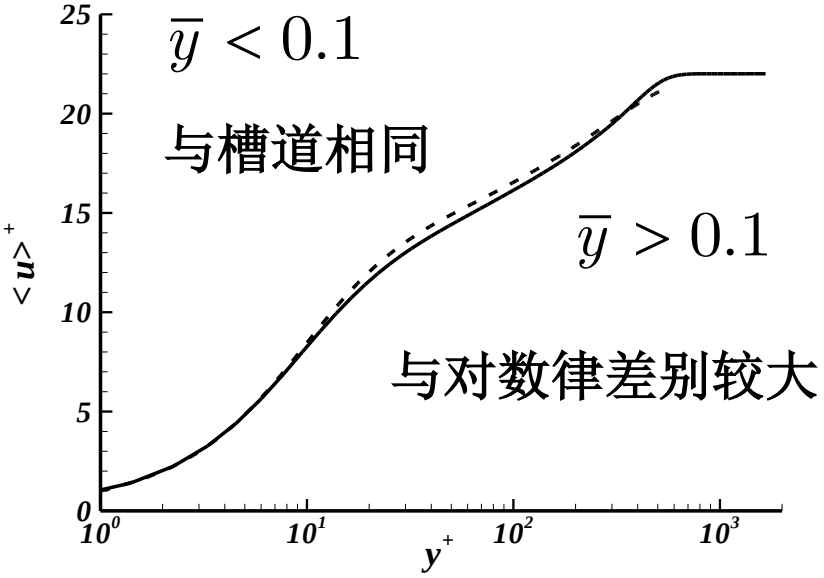
平均速度剖面

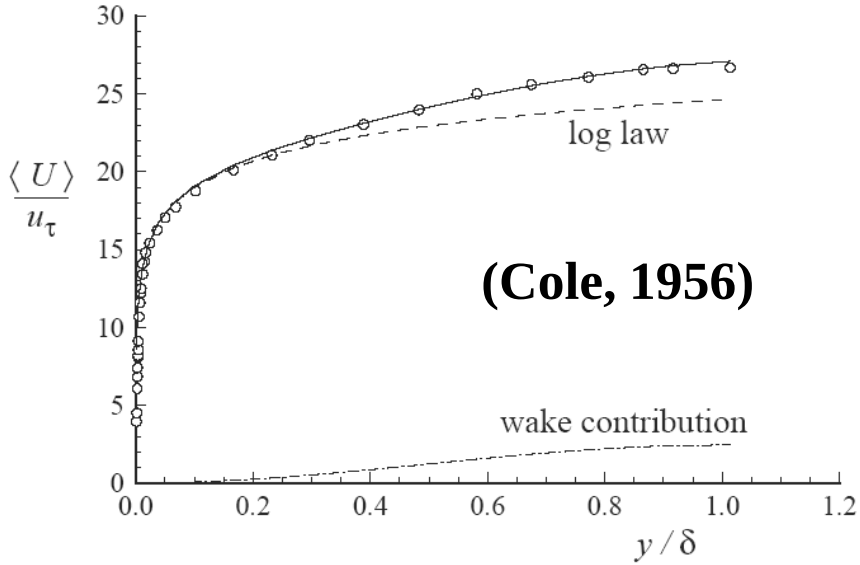




§6.4 壁湍流的统计特性

平均速度分布的壁面律





$$\frac{\langle u \rangle}{u_\tau} = f_w\left(\frac{y}{\delta_\nu}\right) + \frac{\Pi}{\kappa} w\left(\frac{y}{\delta}\right)$$

$$w\left(\frac{y}{\delta}\right) = 2\sin^2\left(\frac{\pi}{2}\frac{y}{\delta}\right)$$

§6.4 壁湍流的统计特性

卡门动量积分方程

$$U\frac{\partial U}{\partial x}+V\frac{\partial U}{\partial y}=U_e\frac{dU_e}{dx}+\nu\frac{\partial^2 U}{\partial y^2}-\frac{\partial}{\partial y}\left\langle u'v'\right\rangle \qquad \frac{\partial U}{\partial x}+\frac{\partial V}{\partial y}=0$$

$$-\frac{1}{\rho}\frac{\partial \tau}{\partial y}=\frac{\partial}{\partial x}\left[U\left(U_e-U\right)\right]+\frac{\partial}{\partial y}\left[V\left(U_e-U\right)\right]+\left(U_e-U\right)\frac{dU_e}{dx}$$

$\int_0^\infty (\ )\,dy$

$$\Longrightarrow \quad \frac{1}{\rho}\tau_w=\frac{d}{dx}\left[U_e^2\theta\right]+\delta^* U_e\frac{dU_e}{dx},\quad \tau_w(x)=\mu\left.\frac{\partial U}{\partial y}\right|_{y=0}$$

$$\Longrightarrow \quad \frac{c_f}{2}=\frac{\tau_w}{\rho U_e^2}=\frac{d\theta}{dx}+\frac{1}{U_e}\frac{dU_e}{dx}\left(2\theta+\delta^*\right)$$

平板零压梯度的情况： $\dfrac{dU_e}{dx}=0 \Longrightarrow \dfrac{c_f}{2}=\dfrac{\tau_w}{\rho U_e^2}=\dfrac{d\theta}{dx}$

§6.4 壁湍流的统计特性

动量积分关系应用

$$U^+=\frac{U}{u_\tau},\quad y^+=\frac{yu_\tau}{\nu},\quad \eta=\frac{y}{\delta}$$

$$U^+=\frac{1}{\kappa}\ln\left(y^+\right)+B+\frac{2\Pi}{\kappa}\sin^2\left(\frac{\pi}{2}\eta\right)$$

入口处的边界层厚度

$$\Longrightarrow\quad \frac{\delta}{\delta_0}=\frac{\lambda}{\mathrm{Re}}\exp\left[\kappa\left(\lambda-B\right)-2\Pi\right],\qquad \lambda\equiv\frac{U_e}{u_\tau}=\sqrt{\frac{2}{c_f}},\qquad \mathrm{Re}=\frac{U_e\delta_0}{\nu}$$

$$\Longrightarrow\quad \frac{U}{U_e}=1+\frac{1}{\kappa\lambda}\left\{\ln\eta+2\Pi\left[\sin^2\left(\frac{\pi}{2}\eta\right)-1\right]\right\}$$

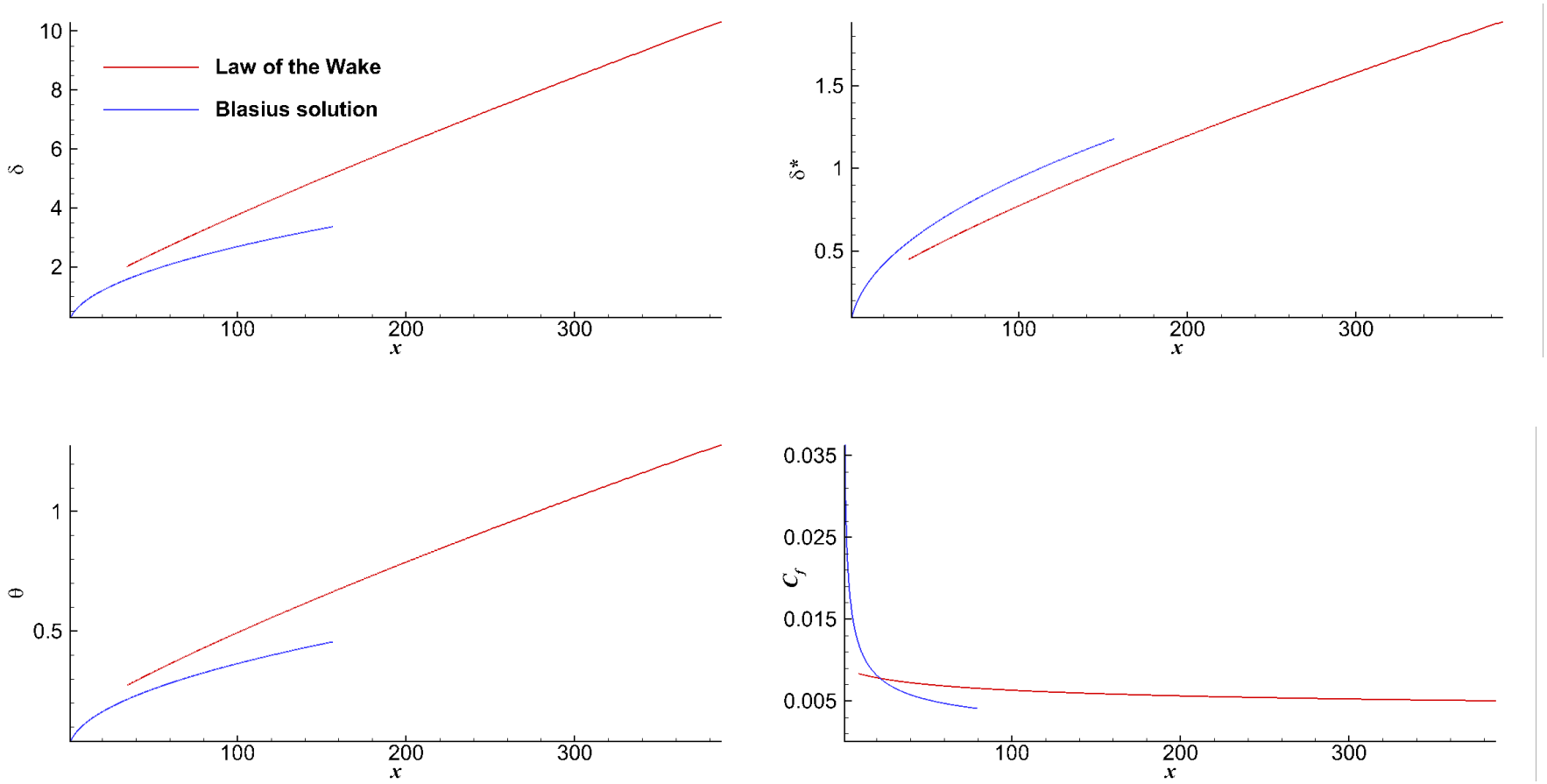
$$\Longrightarrow\quad \delta^*=\left(\frac{1+\Pi}{\kappa\lambda}\right)\delta$$

（自行推导！）

$$\theta=\left\{\frac{1+\Pi}{\kappa\lambda}-\frac{1}{\kappa^2\lambda^2}\left[2+2\Pi\left(\frac{\mathrm{Si}\pi}{\pi}+1\right)+\frac{3}{2}\Pi^2\right]\right\}\delta,\quad \mathrm{Si}\left(x\right)=\int_0^x\frac{\sin t}{t}dt$$

由动量积分关系 $\mathrm{d}x=\lambda^2\mathrm{d}\theta$ 可以得到边界层厚度和壁面摩擦系数沿流向的分布

§6.4 壁湍流的统计特性



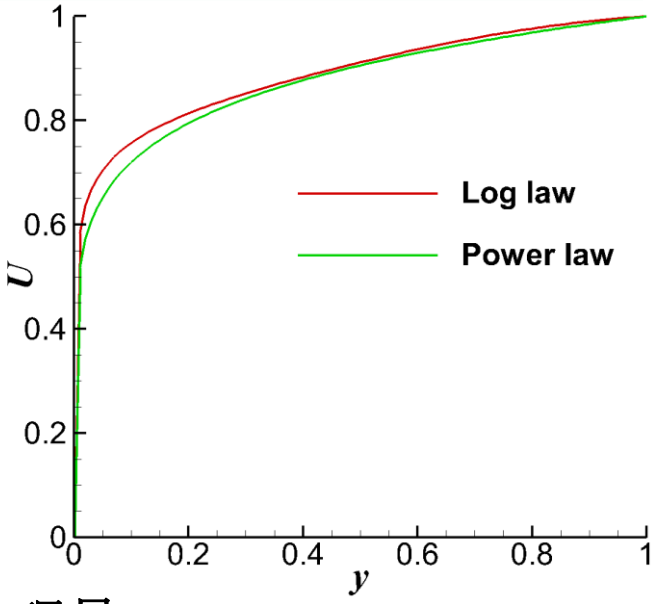
湍流边界层的厚度增长远大于层流情况
壁面摩擦力系数的衰减远小于层流情况

§6.4 壁湍流的统计特性

湍流边界层的时均速度分布除了用对数律，工程估算中常用幂次律近似

$$\frac{U}{U_e}=\left(\frac{y}{\delta}\right)^{1/n}$$

Re_x	10^6~2×10^7	2×10^7~2×10^8	2×10^8~10^{10}
n	7	8	9

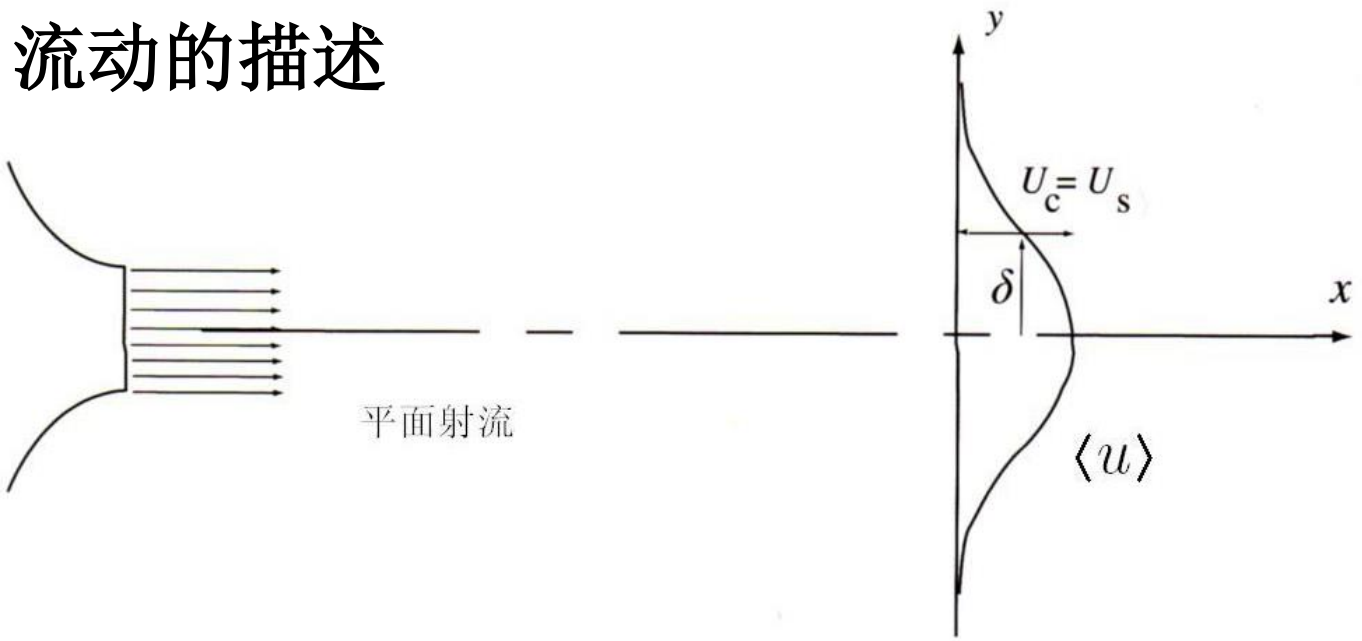


有了速度分布，就可以进一步计算其它的有关物理量。

（自行推导！）

§6.5 自由切变湍流的统计特性

一、流动的描述



特征速度

$$U_c(x) = \frac{1}{2}\langle u(x,0)\rangle$$

特征宽度（半宽度）

$$\frac{1}{2}\langle u(x,0)\rangle = \langle u(x,\pm\delta(x))\rangle$$

§6.5 自由切变湍流的统计特性

二、自由切变湍流的边界层近似（平面流动为例）

$$\frac{\partial U}{\partial x}+\frac{\partial V}{\partial y}=0$$

$$U\frac{\partial U}{\partial x}+V\frac{\partial U}{\partial y}=-\frac{1}{\rho}\frac{\partial P}{\partial x}+\nu\left(\frac{\partial^2 U}{\partial x^2}+\frac{\partial^2 U}{\partial y^2}\right)-\frac{\partial\left\langle u'^2\right\rangle}{\partial x}-\frac{\partial\left\langle u'v'\right\rangle}{\partial y}$$

$$U\frac{\partial V}{\partial x}+V\frac{\partial V}{\partial y}=-\frac{1}{\rho}\frac{\partial P}{\partial y}+\nu\left(\frac{\partial^2 V}{\partial x^2}+\frac{\partial^2 V}{\partial y^2}\right)-\frac{\partial\left\langle u'v'\right\rangle}{\partial x}-\frac{\partial\left\langle v'^2\right\rangle}{\partial y}$$

流场的缓变特性　　$V\sim\varepsilon U$　　$\partial/\partial x\sim\varepsilon\,\partial/\partial y$

$$U\frac{\partial U}{\partial x}+V\frac{\partial U}{\partial y}=-\frac{1}{\rho}\frac{\mathrm{d}p_\infty}{\mathrm{d}x}+\nu\frac{\partial^2 U}{\partial y^2}-\frac{\partial\left\langle u'v'\right\rangle}{\partial y}$$

§6.5 自由切变湍流的统计特性

三、自由切变湍流的自相似性

任意物理量 $Q(x,y)$　　定义特征量　$Q_0(x)$　　$\delta(x)$

如果　$\dfrac{Q(x,y)}{Q_0(x)}=\tilde{Q}(\xi)$　　$\xi\equiv\dfrac{y}{\delta(x)}$

自相似

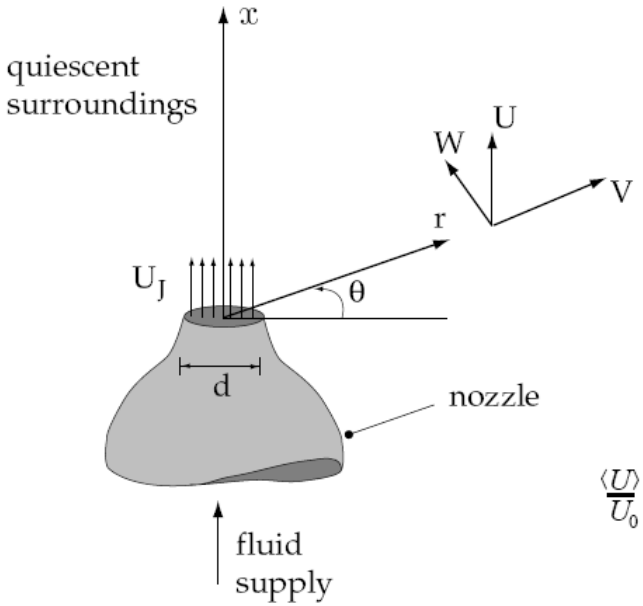
偏微分方程

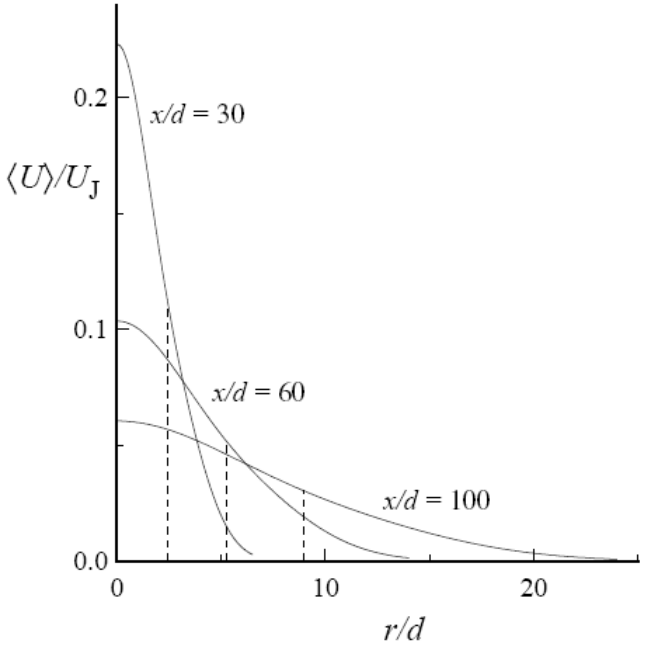
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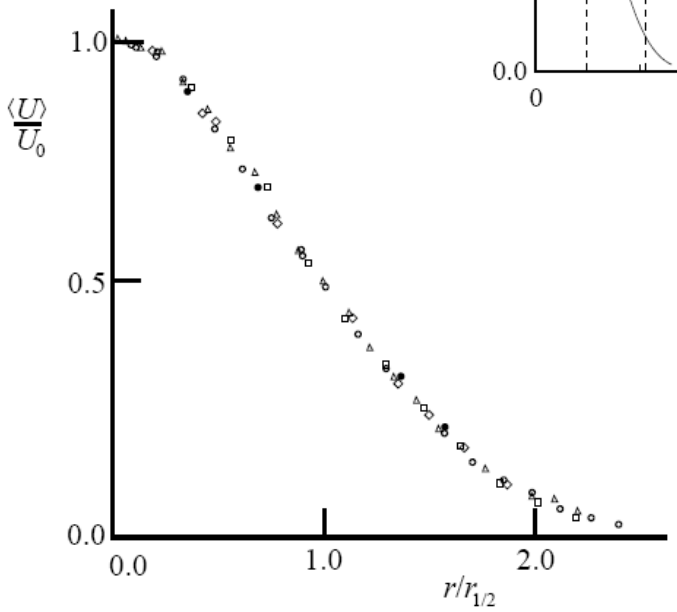
常微分方程

§6.5 自由切变湍流的统计特性

圆射流的自相似性







§6.5 自由切变湍流的统计特性

圆射流的相似性解

$$\frac{\partial U}{\partial x}+\frac{1}{r}\frac{\partial\left(rV\right)}{\partial r}=0$$

$$U\frac{\partial U}{\partial x}+V\frac{\partial U}{\partial r}=\frac{\nu}{r}\frac{\partial}{\partial r}\left(r\frac{\partial U}{\partial r}\right)-\frac{1}{r}\frac{\partial}{\partial r}\left(r\left\langle u'v'\right\rangle\right)$$

连续方程

雷诺应力远大于粘性应力，动量方程略去粘性项，并乘以r

$$rU\frac{\partial U}{\partial x}+rV\frac{\partial U}{\partial r}=-\frac{\partial}{\partial r}\left(r\left\langle u'v'\right\rangle\right)$$

$$rV\frac{\partial U}{\partial r}=\frac{\partial}{\partial r}\left(rVU\right)-U\frac{\partial rV}{\partial r}$$

$$=\frac{\partial}{\partial r}\left(rVU\right)+rU\frac{\partial U}{\partial x}$$

$$\frac{\partial rUU}{\partial x}=-\frac{\partial}{\partial r}\left(rUV+r\left\langle u'v'\right\rangle\right)$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\int_0^\infty rUU\mathrm{d}r=-\left(rUV+r\left\langle u'v'\right\rangle\right)_0^\infty=0$$

平均动量通量守恒

$$\dot{M}(x)=\int_0^\infty 2\pi r\rho U^2\mathrm{d}r \qquad \frac{\mathrm{d}}{\mathrm{d}x}\dot{M}(x)=0$$

§6.5 自由切变湍流的统计特性

相似变换 $U=U_0(x)f(\xi)$ $\langle u'v'\rangle=U_0^2(x)g(\xi)$ $\xi=\dfrac{r}{r_{1/2}(x)}$

$$\frac{\partial rV}{\partial r}=-r\frac{\partial U}{\partial x}=-r\left(\frac{\mathrm{d}U_0}{\mathrm{d}x}f+U_0\frac{\mathrm{d}f}{\mathrm{d}\xi}\frac{\partial\xi}{\partial x}\right)$$

$$=-r\left[\frac{\mathrm{d}U_0}{\mathrm{d}x}f+U_0\frac{\mathrm{d}f}{\mathrm{d}\xi}\left(-\frac{r}{r_{1/2}^2}\frac{\mathrm{d}r_{1/2}}{\mathrm{d}x}\right)\right]=-r_{1/2}\frac{\mathrm{d}U_0}{\mathrm{d}x}\xi f+U_0\frac{\mathrm{d}r_{1/2}}{\mathrm{d}x}\xi^2\frac{\mathrm{d}f}{\mathrm{d}\xi}$$

$$\frac{\partial U}{\partial x}+\frac{1}{r}\frac{\partial(rV)}{\partial r}=0$$

$$V=\frac{1}{r}\int_0^r\left(-r_{1/2}\frac{\mathrm{d}U_0}{\mathrm{d}x}\xi f+U_0\frac{\mathrm{d}r_{1/2}}{\mathrm{d}x}\xi^2\frac{\mathrm{d}f}{\mathrm{d}\xi}\right)\mathrm{d}r=\frac{1}{r}\int_0^\xi\left(-r_{1/2}\frac{\mathrm{d}U_0}{\mathrm{d}x}\xi f+U_0\frac{\mathrm{d}r_{1/2}}{\mathrm{d}x}\xi^2\frac{\mathrm{d}f}{\mathrm{d}\xi}\right)r_{1/2}\mathrm{d}\xi$$

$$=-\frac{r_{1/2}^2}{r}\frac{\mathrm{d}U_0}{\mathrm{d}x}\int_0^\xi\xi f\mathrm{d}\xi+U_0\frac{r_{1/2}}{r}\frac{\mathrm{d}r_{1/2}}{\mathrm{d}x}\int_0^\xi\xi^2\frac{\mathrm{d}f}{\mathrm{d}\xi}\mathrm{d}\xi \longrightarrow \xi^2f-2\int_0^\xi\xi f\mathrm{d}\xi$$

$$=U_0\frac{\mathrm{d}r_{1/2}}{\mathrm{d}x}\xi f-\left(r_{1/2}\frac{\mathrm{d}U_0}{\mathrm{d}x}+2U_0\frac{\mathrm{d}r_{1/2}}{\mathrm{d}x}\right)\frac{1}{\xi}\int_0^\xi\xi f\mathrm{d}\xi$$

$$V=U_0\left[\frac{\mathrm{d}r_{1/2}}{\mathrm{d}x}\xi f-\left(\frac{r_{1/2}}{U_0}\frac{\mathrm{d}U_0}{\mathrm{d}x}+2\frac{\mathrm{d}r_{1/2}}{\mathrm{d}x}\right)\frac{1}{\xi}\int_0^\xi\xi f\mathrm{d}\xi\right]$$

§6.5 自由切变湍流的统计特性

$$U = U_0(x) f(\xi) \qquad V = U_0\left[\frac{\mathrm{d}r_{1/2}}{\mathrm{d}x}\xi f - \left(\frac{r_{1/2}}{U_0}\frac{\mathrm{d}U_0}{\mathrm{d}x} + 2\frac{\mathrm{d}r_{1/2}}{\mathrm{d}x}\right)\frac{1}{\xi}\int_0^\xi \xi f \mathrm{d}\xi\right] \qquad \langle u'v'\rangle = U_0^2(x)\, g(\xi) \qquad \xi = \frac{r}{r_{1/2}(x)}$$

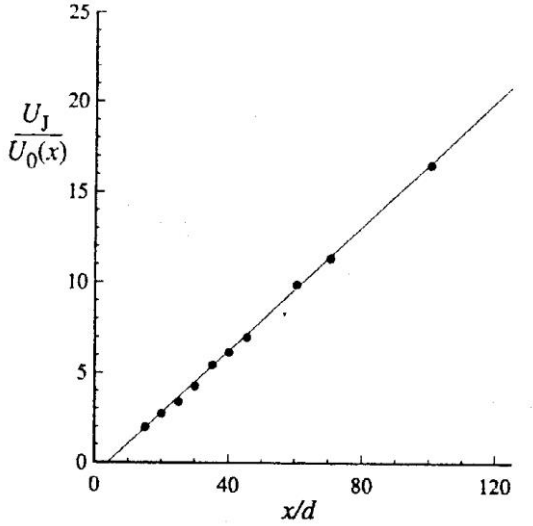
$$U\frac{\partial U}{\partial x} + V\frac{\partial U}{\partial r} = -\frac{1}{r}\frac{\partial}{\partial r}\left(r\langle u'v'\rangle\right)$$

$$\left(\frac{r_{1/2}}{U_0}\frac{\mathrm{d}U_0}{\mathrm{d}x}\right)\xi f^2 - \left(\frac{r_{1/2}}{U_0}\frac{\mathrm{d}U_0}{\mathrm{d}x} + 2\frac{\mathrm{d}r_{1/2}}{\mathrm{d}x}\right)\frac{\mathrm{d}f}{\mathrm{d}\xi}\int_0^\xi \xi f \mathrm{d}\xi = -\frac{\mathrm{d}}{\mathrm{d}\xi}\left(\xi g\right)$$

$\Rightarrow$ $\dfrac{\mathrm{d}r_{1/2}}{\mathrm{d}x} = S$ $\qquad r_{1/2} \propto x$

$$\dot{M}(x) = \int_0^\infty 2\pi r\rho U^2 \mathrm{d}r = 2\pi\rho U_0^2 r_{1/2}^2 \int_0^\infty \xi f^2 \mathrm{d}\xi$$

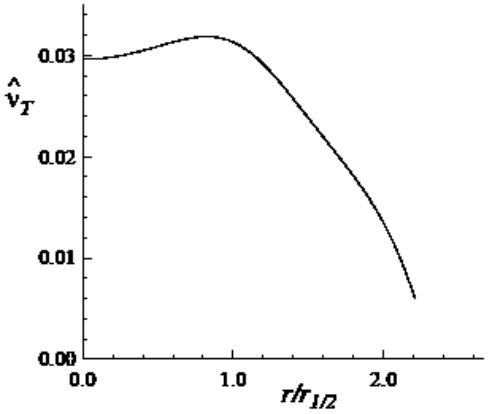
$\dfrac{\mathrm{d}}{\mathrm{d}x}\dot{M}(x) = 0$ $\Rightarrow$ $U_0 \propto \dfrac{1}{r_{1/2}} \propto \dfrac{1}{x}$



§6.5 自由切变湍流的统计特性

涡粘假设 $\langle u'v' \rangle = -\nu_T \dfrac{\partial U}{\partial r}$

$U_0 r_{1/2} = const$



自相似 $\nu_T(x,r) = U_0(x)\, r_{1/2}(x)\, \hat{\nu}_T(\xi)$

射流核心区 $\nu_T = \text{const.}$

$$U\frac{\partial U}{\partial x} + V\frac{\partial U}{\partial r} = \frac{\nu_T}{r}\frac{\partial}{\partial r}\left(r\frac{\partial U}{\partial r}\right) \qquad \frac{\partial U}{\partial x} + \frac{1}{r}\frac{\partial (rV)}{\partial r} = 0$$

方程组封闭

定义流函数 $U = \dfrac{1}{r}\dfrac{\partial \psi}{\partial r}$ $V = -\dfrac{1}{r}\dfrac{\partial \psi}{\partial x}$

连续方程自动满足

定义相似变量 $\xi = \dfrac{r}{x}$

$$\psi = \int rU\mathrm{d}r = \int x\xi U_0 f x\mathrm{d}\xi = x^2 U_0 \int \xi f \mathrm{d}\xi$$

$$\psi = \nu_T x F(\xi)$$

$x^2 U_0 \sim x$

§6.5 自由切变湍流的统计特性

$$\psi = \nu_T x F(\xi) \qquad U = \frac{1}{r}\frac{\partial \psi}{\partial r} \qquad V = -\frac{1}{r}\frac{\partial \psi}{\partial x}$$

$$U\frac{\partial U}{\partial x} + V\frac{\partial U}{\partial r} = \frac{\nu_T}{r}\frac{\partial}{\partial r}\left(r\frac{\partial U}{\partial r}\right) \qquad \xi = \frac{r}{x}$$

$$U = \frac{1}{r}\nu_T x F'\frac{\partial \xi}{\partial r} = \frac{\nu_T}{x}\frac{F'}{\xi} \qquad V = -\frac{\nu_T}{r}\left(F + xF'\frac{\partial \xi}{\partial x}\right) = \frac{\nu_T}{x}\left[F' - \frac{1}{\xi}F\right]$$

$$\frac{FF'}{\xi^2} - \frac{F'^2}{\xi} - \frac{FF''}{\xi} = F''' - \frac{F''}{\xi} + \frac{F'}{\xi^2}$$

$$\Downarrow$$

$$\left(-\frac{FF'}{\xi}\right)' = \left(F'' - \frac{F'}{\xi}\right)' \Rightarrow -\frac{FF'}{\xi} = F'' - \frac{F'}{\xi} + A$$

$r = 0 \qquad V = 0 \qquad \dfrac{\partial U}{\partial r} = 0$

$$\Downarrow$$

$$F(0) = F'(0) = F''(0) = 0$$

$$A = 0 \qquad FF' = F' - \xi F''$$

$$\Downarrow$$

$$\left(\frac{1}{2}F^2\right)' = 2F' - \left(\xi F'\right)' \Rightarrow \frac{1}{2}F^2 = 2F - \xi F' \Rightarrow \frac{1}{2F - \frac{1}{2}F^2}\mathrm{d}F = \frac{1}{\xi}\mathrm{d}\xi$$

$$\Rightarrow \frac{1}{2}\ln\left(\frac{F}{4-F}\right) = \ln\xi + C \qquad F = \frac{4a\xi^2}{1 + a\xi^2}$$

§6.5 自由切变湍流的统计特性

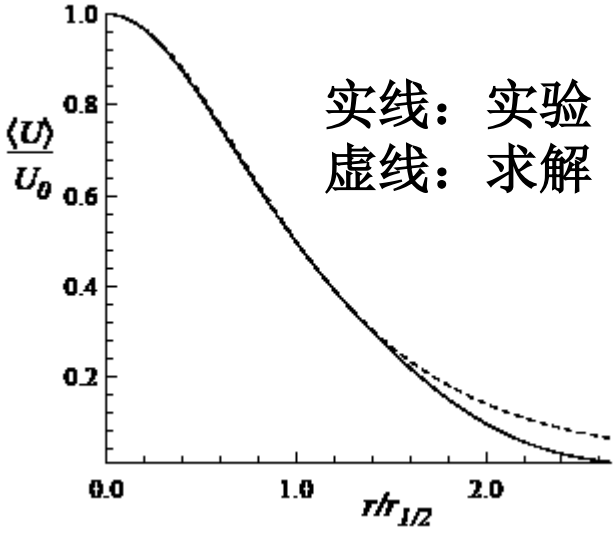
$$F=\frac{4a\xi^2}{1+a\xi^2} \qquad U=\frac{\nu_T}{x}\frac{F'}{\xi} \qquad U=U_0 f(\xi)$$

$$U=\frac{8a\nu_T}{x}\frac{1}{\left(1+a\xi^2\right)^2} \qquad U_0(x)=\frac{8a\nu_T}{x} \qquad f(\xi)=\frac{1}{\left(1+a\xi^2\right)^2}$$

确定常数a

$$r_{1/2}=Sx \qquad r=r_{1/2} \qquad \xi=\frac{r_{1/2}}{x}=S$$

$$f(S)=\frac{1}{\left(1+aS^2\right)^2}=\frac{1}{2} \Rightarrow a=\frac{\sqrt{2}-1}{S^2}$$



$$\nu_T(x,r)=U_0(x)r_{1/2}(x)\hat{\nu}_T(\xi)$$

$$\hat{\nu}_T=\frac{S}{8\left(\sqrt{2}-1\right)}$$

$$U_0(x)=\frac{8a\nu_T}{x}$$

§6.5 自由切变湍流的统计特性

自由切变湍流统计特性的比较

特征宽度 $\delta \sim x^p$ **特征速度** $U_0 \sim x^q$

	层流		湍流	
	p	q	p	q
轴对称射流	1	−1	1	−1
平面射流	2/3	−1/3	1	−1/2
轴对称尾流	1/2	−1	1/3	−2/3
平面尾流	1/2	−1/2	1/2	−1/2
平面混合层	1/2	0	1	0